

RL Portfolio Optimization v3

Phase 1: DCC-GARCH GAT + HMM Regime + Expanded Universe

Assets: AAPL | MSFT | GOOGL | AMZN | TSLA | SPY | TLT | GLD | IWM | EFA

Train: 2000-2019 | Val: 2020-2022 | Test: 2023-2024

Algorithm: PPO + CVaR (ALM) | N_ACTIONS=11

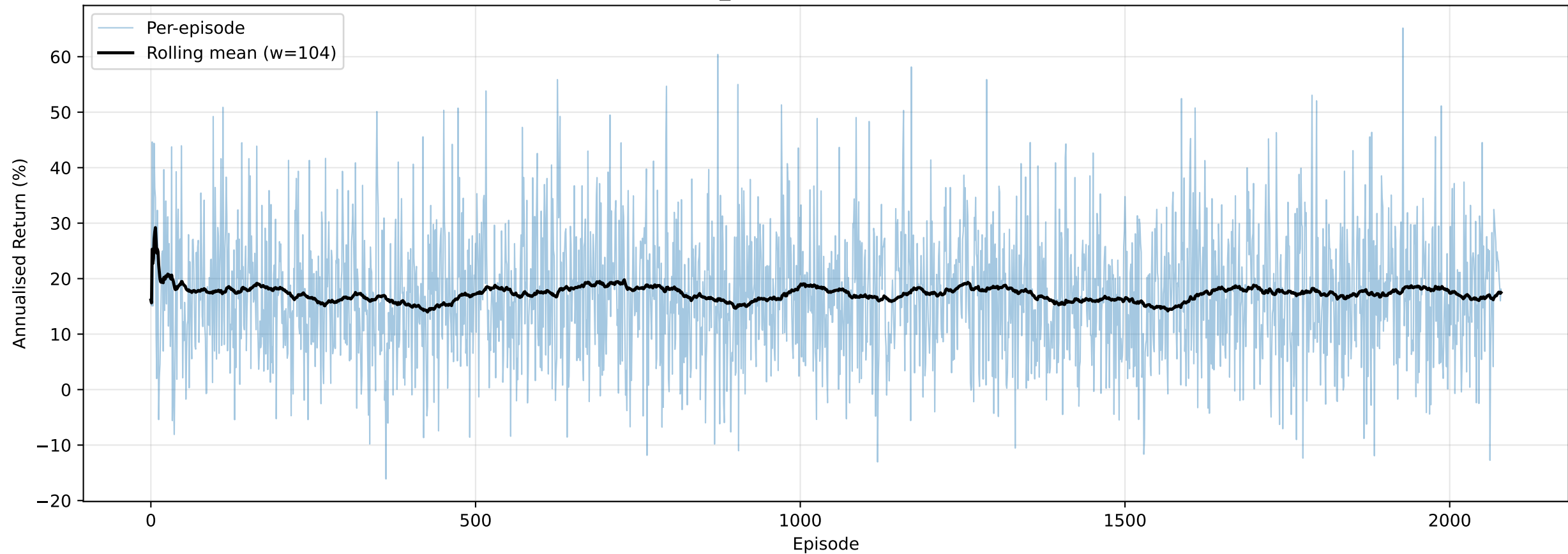
Test Results (2023-2024)

Experiment	Cum.Return	Sharpe	MaxDrawdown	CVaR_0.95	Turnover
PPO_CVaR	+70.79%	1.667	-10.54%	-0.0186	0.0376
EqualWeight	+83.11%	1.718	-11.44%	-0.0209	0.0000

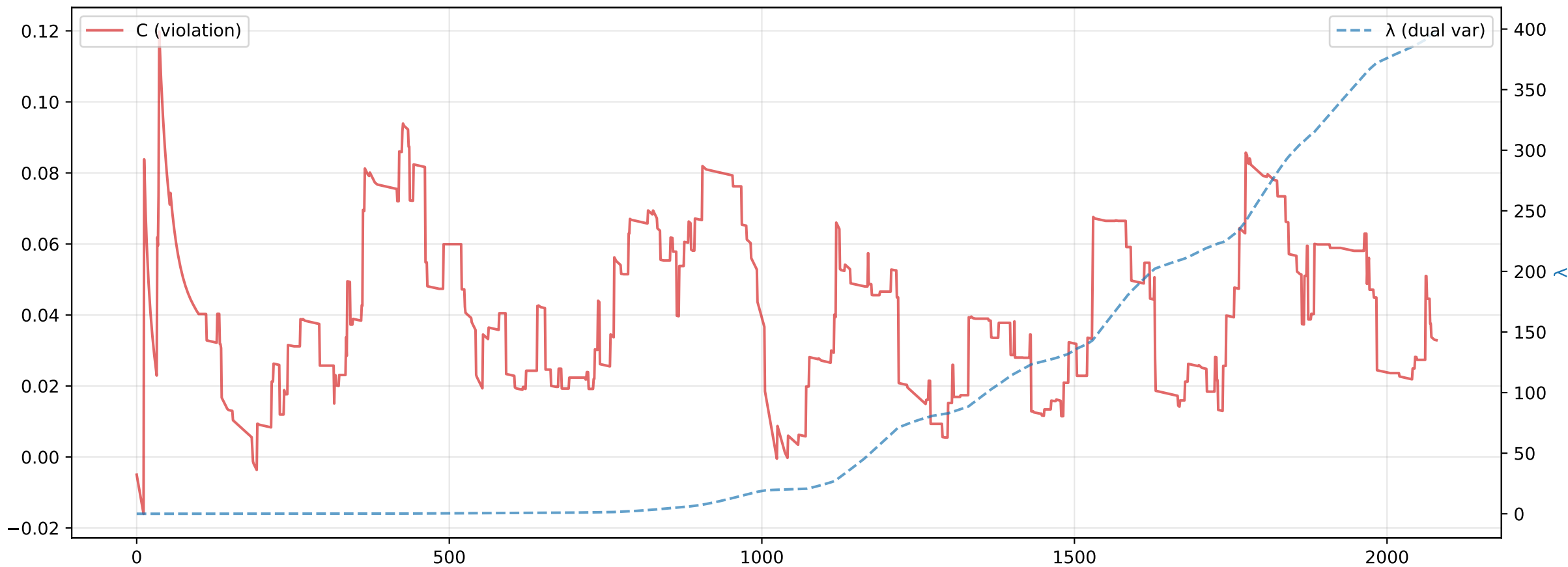
Cumulative Return — Test Set (2023-2024)



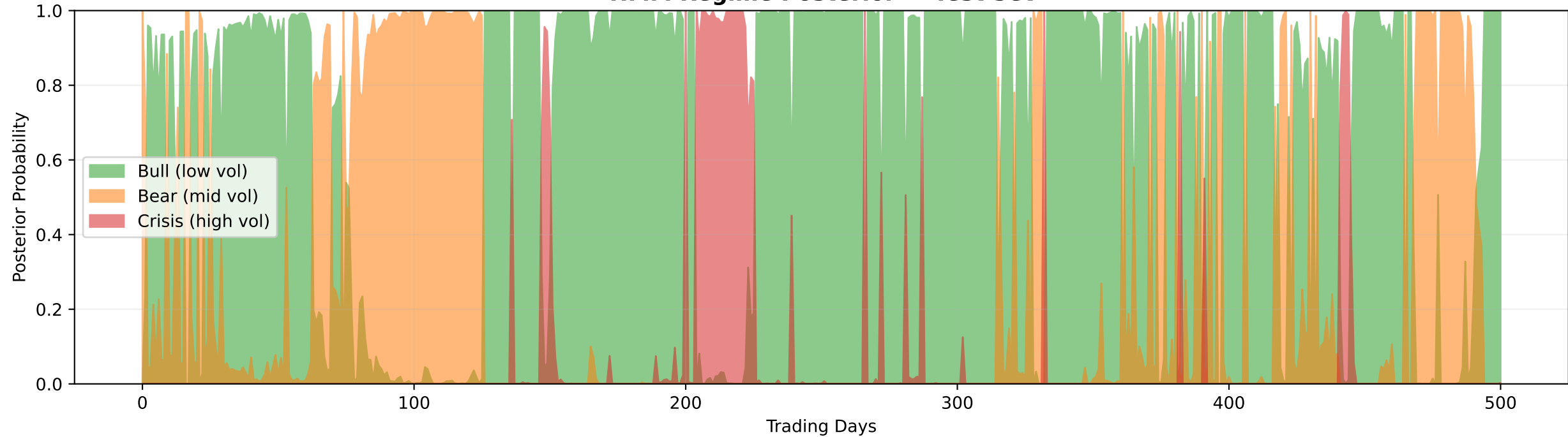
PPO_CVaR Training Curve



CVaR Constraint — Violation & Dual Variable

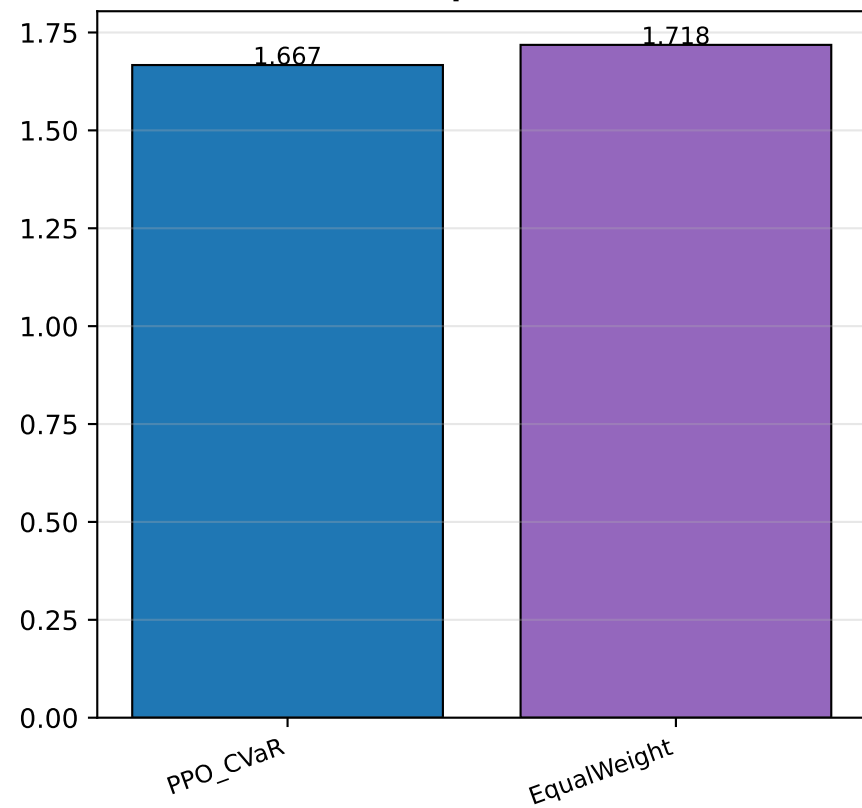


HMM Regime Posterior — Test Set

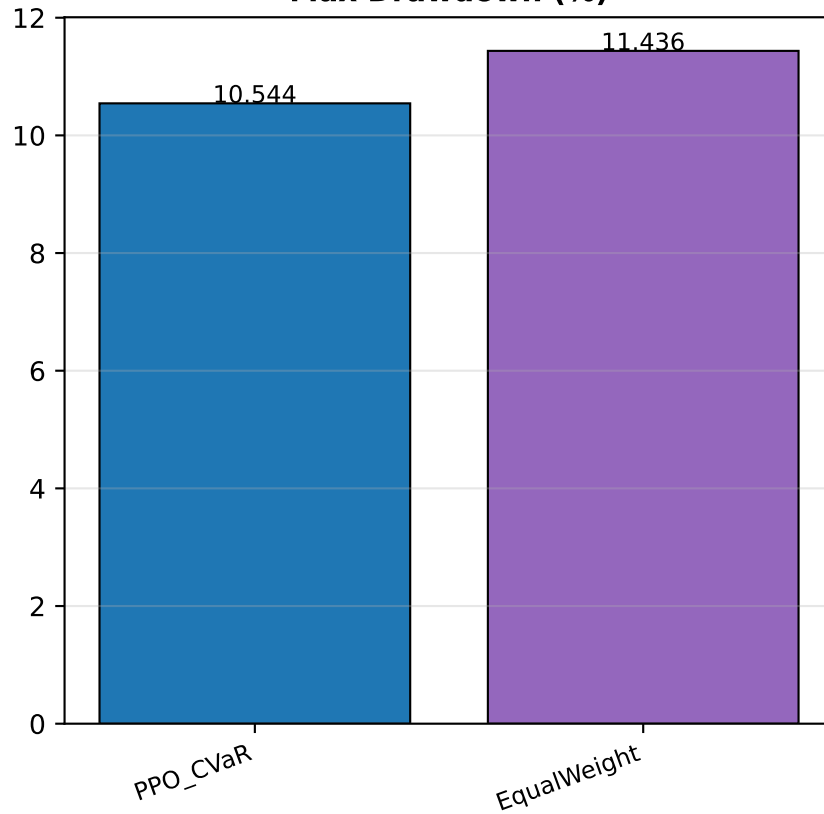


Test Set Metrics (2023-2024)

Sharpe Ratio



Max Drawdown (%)



Avg Turnover

